



Subdaily PM_{2.5} exposure and cardiorespiratory risks: data and findings from Southern California, 2018–2020

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Abstract

As hourly PM_{2.5} measurements become more accessible, health impacts from subdaily exposures can be evaluated to develop health guidance. We obtained hourly PM_{2.5} concentrations covering Southern California from May 2018 through March 2020 and daily emergency department visits (EDVs) for cardiorespiratory-related conditions at ZIP Code Tabulation Area (ZCTA) levels. ZCTAs were aggregated into 35 clusters based on similar geographic and sociodemographic features. Daily exceedance concentration hours (DECH) above 9, 12, and 15 µg/m³, daily maximum, and average PM_{2.5} concentrations were calculated for each cluster-day. Two-stage time-series analyses were conducted to estimate excess risks of daily EDVs. DECH metrics exhibited the same direction but smaller effects on cardiovascular and respiratory EDVs compared to daily average metrics. Excess risks for cardiovascular EDVs were 1.77% (95% CI: 1.20, 2.34), 1.04% (0.61, 1.47), and 2.67% (1.98, 3.37) per interquartile range increase of DECH-9, DECH-12, and daily average PM_{2.5} during 7-day lag period, respectively. Excess risks of respiratory EDVs increased by 6.34% (4.25, 8.48), 4.39% (2.85, 5.95), and 6.61% (4.78, 8.47) per IQR increase of DECH-9, DECH-12, and daily average PM_{2.5} during a 3-day lag period, respectively. Elevated excess risks were observed among older adults (65+), children (0–17), and low-poverty neighborhoods on both subdaily and daily metrics. In summary, subdaily PM_{2.5} exposures above the current standards exhibited excess risks in cardiorespiratory-related EDVs but no greater than those derived from the daily average metric. Health guidance based on the daily average metric provides sensible protection to the public in Southern California.

Keywords PM_{2.5} · Hourly concentration · Daily exceedance concentration hours (DECH) · Cardiovascular · Respiratory

Introduction

Ambient exposure to particulate matter with a diameter of 2.5 microns or less (PM_{2.5}) has been associated with adverse respiratory, cardiovascular, and metabolic health outcomes (Atkinson et al. 2014; Li et al. 2016; Shah et al. 2015; USEPA 2019). Monitoring of ambient PM_{2.5} with hourly and finer time resolution has become

more common, enabling the ability to examine exposures within a day, i.e., subdaily exposure. Subdaily exposure metrics capture exposure in a single hour, exposure during commute hours, daytime average exposure, or the number of hours exceeding a certain limit, so that the peak exposure from short-term events, e.g., fires, industrial operations, fireworks, etc., can be examined. Subdaily exposure may unveil adverse health outcomes that are not observed or that may be attenuated when only measuring at the daily level. However, research on subdaily exposure to PM_{2.5} is sparse, especially in the United States, and the current ambient air quality standards are based on daily and/or annual concentrations. Interpretation of subdaily data and their practical applications to public health remain uncertain.

Globally, literature investigating subdaily exposure to PM_{2.5} has suggested associations with various health outcomes, such as, asthma, chronic obstructive

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pulmonary disease (COPD), high blood pressure, acute myocardial infarction (MI), and even hampered cognitive performance (Chen et al. 2020; Cheng et al. 2021; Cleland et al. 2022; Li et al. 2019; Lin et al. 2017, 2021; Zhang et al. 2021). For example, studies in Australia demonstrated elevated risk of MI and asthma aggravation in children within several short (1-hour) to moderate (7–12 h) exposure windows of PM_{2.5} (Cheng et al. 2021, 2022). However, very few of such studies were conducted in the U.S.

Furthermore, it is difficult to consolidate these findings, given the wide variety of subdaily exposure metrics used and various classifications of health outcomes. For example, subdaily metrics used in the literature ranged from single-hour concentrations, cumulative hour windows (e.g., 1–2, 2–6, 7–12 h prior), to aggregate metrics such as daily concentration exceedance hours (DECH). The health outcomes varied from subclinical changes in biomarkers, medication use, outpatient clinic visits, urgent care use, ambulance dispatch, emergency department visits (EDVs), hospitalizations, to mortality. Therefore, the evidence for each individual health outcome is limited and insufficient to build concentration-response relationships for subdaily exposures. Based on the existing limited literature, the United States Environmental Protection Agency's (USEPA) 2019 Integrated Science Assessment (ISA) for Particulate Matter (PM) concluded that subdaily exposures to PM_{2.5} were similar in effect when compared to daily average exposure, with inconsistency across metrics (Jiao et al. 2020; USEPA 2019; Zhang et al. 2021).

Previous research has also suggested differential health burdens of PM_{2.5} across various socioeconomic standing and race/ethnicity groups. Some studies have observed stronger associations between PM_{2.5} and cardiorespiratory outcomes among people of various age groups living in lower socioeconomic status (SES) neighborhoods, while others have reported no modification by neighborhood SES (Alexeeff et al. 2023; Cakmak et al. 2016; Chi et al. 2016; Goodman et al. 2017; Wang et al. 2017). These relationships have not been extensively explored with subdaily exposure metrics (Mehta et al. 2021).

To fill these knowledge gaps, we investigated associations of subdaily exposure to PM_{2.5} and cardiorespiratory EDVs in Southern California to evaluate if subdaily metrics of PM_{2.5} exposure have similar health effects to daily metrics. We also evaluated if there are heterogeneous risks across sociodemographic subgroups by age group, race/ethnicity, and neighborhood poverty level, to identify disproportionate impact across populations.

Methods

Air pollutants

We procured hourly estimations of PM_{2.5} (µg/m³), ozone (O₃, ppb), and nitrogen dioxide (NO₂, ppb) concentration for the period May 2018 - March 2020 from the South Coast Air Quality Management District (South Coast AQMD). The concentration data was generated using the method used by the South Coast AQMD real-time air quality index map (Schulte et al. 2020). This method combines data from multiple sources: (1) measurements from the South Coast AQMD network of regulatory monitors, (2) PM_{2.5} data from hundreds of PurpleAir sensors (PurpleAir 2023), and (3) modeled ozone and PM_{2.5} fields from the National Air Quality Forecast Capability Community Multiscale Air Quality model (CMAQ) issued by the National Oceanic and Atmospheric Administration (NOAA) (Lee et al. 2017). The measurements and model data are combined using a kriging technique that takes into account the uncertainty of each data source. The resulting hourly concentration data were available at approximate 5 km-by-5 km horizontal grid resolution at the surface and covered the South Coast AQMD jurisdiction - a region including the entirety of Orange County, Los Angeles County except for the High Desert, Riverside County except for the Palo Verde Valley region, and the southwestern portion of San Bernardino County. We assigned the hourly concentrations to ZIP Code Tabulation Areas (ZCTAs) based on the grid that corresponded to their population-weighted centroid. Although PM_{2.5} was our main exposure of interest, O₃ and NO₂ were included as co-pollutants in the sensitivity analyses. Their correlations were examined using Spearman correlation analysis, to ensure that they can be included in the co-pollutant model without having the collinearity issue.

In the dataset, there were a small portion of negative values (PM_{2.5}: 1.02% of observations; O₃: 0.98%; NO₂: 0.10%), which is typical for datasets containing low measurements with uncertainty. We replaced all values below the 5th percentile of negative values (PM_{2.5}: -4.68 µg/m³; also for O₃: -3.75 ppb and NO₂: -2.19 ppb) with -1 µg/m³ or ppb. Extremely high PM_{2.5} concentrations (0.001% of the dataset had PM_{2.5} > 500 µg/m³, max 1000 µg/m³), which were likely due to short-term weather events (e.g., dust storms) or local sources (e.g., fireworks or wildfires), were retained in the dataset. Concentration data farther than 140 km from a valid regulatory monitor measurement were removed because these data were subject to higher uncertainty. Missing data were minimal (PM_{2.5}: 0.09%; O₃: 0.04%; NO₂: 0.05%) on an hourly basis. While there were some missing hourly estimates for New Years Eve/Day of

2019/2020, we kept them in the analyses because data completeness was greater than 80%.

Daily exceedance concentration hours (DECH), a novel measure that sums the amount of $PM_{2.5}$ exposure over a certain hourly threshold, for the entire day, were adopted as primary subdaily metrics in this study (Lin et al. 2018). For example, DECH-12 refers to the sum of the concentration-hours over $12 \mu\text{g}/\text{m}^3$ for a day: one hour of $50 \mu\text{g}/\text{m}^3$ would contribute $38 \mu\text{g}/\text{m}^3 \cdot \text{hour}$. DECH-9, 12, 15 were calculated from hourly concentration for each geographic unit. The thresholds of 9 and $12 \mu\text{g}/\text{m}^3$ were selected to correspond to the USEPA's current and past annual standard for $PM_{2.5}$, respectively (USEPA 2024). The threshold of $15 \mu\text{g}/\text{m}^3$ was selected based on the daily average air concentration guideline suggested by the World Health Organization (WHO) (WHO 2021). Daily (24-hour) maximum and daily (24-hour) average were also calculated from hourly concentrations.

Health outcomes

Daily individual patient EDV data from May 2018 to March 2020 were obtained from the Department of Health Care Access and Information (HCAI) to match our exposure data, with each record corresponding to an individual patient's hospitalization/EDV, complete with visit date, residential ZCTA, ICD-10-CM (International Classification of Diseases, Tenth Revision, Clinical Modification) code of primary diagnosis, age, and race/ethnicity data. The exact hourly time stamp of each visit was not available, and therefore, EDV counts were aggregated to daily levels. This analysis focused on EDVs that had primary diagnosis ICD-10-CM codes of respiratory (acute and other) causes (J00 – J99), and cardiovascular-related causes (I00 – I99). Daily counts of EDV cases were summed for each of the 465 ZCTAs within the South Coast AQMD jurisdiction.

Covariates

Demographic and socioeconomic data were procured from the American Community Survey (ACS) for 2015–2019 as collected by the U.S. Census Bureau for each ZCTA in the study area (accessed Sept 2022). These data included proportion of Black, Hispanics, and household income less than two times the federal poverty level. We used these three covariates in the regionalization (described below), and poverty level was also used in the stratified analyses.

Increases in apparent temperature has been found to be positively associated with EDVs of many health outcomes (Basu et al. 2012); therefore, it was adjusted in all analyses. Ambient temperature and relative humidity data were obtained from the USEPA through the Air Quality System

Data Mart, California Irrigation Management Information System and National Oceanic Atmospheric Administration (NOAA). Each ZCTA was assigned the apparent temperature, a combination of air temperature and relative humidity that approximates humans' perceptions of cold and heat (Basu et al. 2018), for the nearest monitor from its centroid. The three-day rolling average apparent temperature that covered the same day and two days prior were adjusted in the analysis to account for immediate and lagged cold or heat impact; we utilized this time frame to capture the prolonged effects of extreme temperatures based on previous literature (Basu and Samet 2002; Lin et al. 2009; Peng et al. 2008).

Regionalization

Given the sparse daily EDV counts at the ZCTA level, the model stability and convergence were of concern, as well as potentially wide confidence intervals (CI). To increase the population size of the unit while retaining sociodemographic cohesion, we clustered ZCTAs using a multivariable spatial clustering procedure. Clusters were constructed from spatially contiguous ZCTAs (i.e., sharing at least two vertices or points) with similar race/ethnicity composition (proportion of Black and Hispanic residents) and poverty level (percentage of households with income less than two times the federal poverty limit) utilizing the R *spdep* package and SKATER algorithm (Assunção et al. 2006). Non-contiguous ZCTAs (e.g., islands, where a GIS polygon was not connected with surrounding ZCTAs) were reassigned manually using a multinomial logistic regression model matching to the nearest clusters. The procedure was parameterized by specifying the number of clusters in the final assignment and by requiring a minimum population requirement in each cluster. We performed this procedure with trials of different cluster count constraints from 10 to 50, in increments of 5, and different minimum population constraints from $\geq 10,000$ to $\geq 50,000$, in increments of 5,000.

We optimized cluster fit in two ways: minimizing total mean square error (TMSE) as a function of sociodemographic fit and weighted ANOVA F-statistic as a function of $PM_{2.5}$ measures. A minimized TMSE of the sociodemographic variable fit and the maximized ratio of the between-cluster variability to within-cluster variability measured by the ANOVA F-statistic for $PM_{2.5}$ measures were considered as the best model, with the former indicating cluster separation in sociodemographic characteristics and the latter indicating maximal differentiation in $PM_{2.5}$ variability between clusters.

After optimizing cluster fit, we calculated the sum of daily EDV counts, the population-weighted hourly average for each air pollutant metric ($PM_{2.5}$, NO_2 , O_3), the

population-weighted daily average for apparent temperature, and the population-weighted census covariates for each cluster.

ZCTAs that were not tabulated by the U.S. Census were excluded, because demographic data may not have been collected, the population was too small to contribute EDV counts, and/or the population was too far from air quality monitors to have reliable estimates. Further, ZCTAs including Veteran Affairs healthcare facilities, college campuses, and amusement parks were also excluded because of their skewed demographic characteristics and/or insufficient case counts in the HCAI database ($n=10$). The final analytical sample consisted of 455 ZCTAs within the South Coast AQMD jurisdiction.

Statistical analysis

Primary model

We employed a two-stage time-series analyses to estimate the excess risk of daily EDVs associated with various $PM_{2.5}$ exposure metrics, including daily maximum $PM_{2.5}$ concentration and DECH-9, -12, -15 as subdaily exposure metrics. Daily average $PM_{2.5}$ was also examined to serve as a reference and for comparison with existing research.

In the first stage, we used a quasi-Poisson regression with linear distributed lags for $PM_{2.5}$ exposures after controlling for temporal trends and other potential covariates in each cluster. Quasi-Poisson was applied to account for over-dispersed outcome data. In the second stage, individual cluster estimates were combined into an overall estimate using random effects meta-analyses with maximum likelihood estimators.

The first-stage model regressed daily counts of EDVs on each $PM_{2.5}$ metric while also including apparent temperature, a holiday indicator variable, the natural log of a cluster's population, and a natural cubic spline to account for long-term temporal trends (Bell et al. 2008). The model is defined as the following form.

$$\log[E(Y_t)] = \beta_0 + \beta_1 cb(PM_{2.5t}) + \beta_2 sma_{t-3}(Temp_t) + ns(t, df) + \beta_3 weekend_t + \beta_4 holiday_t + \beta_5 \log(pop) + \epsilon$$

Where Y_t represents counts of respiratory or cardiovascular-related EDVs on day t . β_0 is the intercept, $cb(PM_{2.5t})$ is the crossbasis matrix, which is combined matrices of $PM_{2.5}$ and lag time, explaining the pollutant and lag simultaneously, $sma_{t-3}(Temp_t)$ is the rolling average of apparent temperature for the 3 days preceding the EDV outcome, ns is the natural cubic spline to account for long-term trend, $weekend_t$ is representing a binary weekend indicator

variable, $holiday_t$ is the holiday indicator variable, $\log(pop)$ is the natural log of a cluster's population, and the error term ϵ . The holiday indicator variable included New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving, and Christmas Day.

Model deviance residuals over time were used to diagnose model fit and appropriate control for time trends, as was minimizing quasi-Akaike's Information Criterion (qAIC) for all $PM_{2.5}$ metrics and lag lengths. Model parameterization in lag length, degrees of freedom (d.f.) for distributed lags, and d.f. in natural cubic spline for long-term trend were sourced through prior research. Four different distributed lag metrics (1, 3, 7, and 14 days) were considered - using the *dlm* package in R (Gasparrini 2011). We fitted polynomial function with 1 d.f. for 1-day and 3-day lags, 2 d.f. for 7-day lag, and 4 d.f. for 14-day lag, respectively. Given the evidence of linear relationship between $PM_{2.5}$ and cardiorespiratory outcomes, we maintained a linear relationship in the distributed lag models of $PM_{2.5}$. For the natural cubic spline controlling for long-term trends, qAIC decreased with larger d.f.: we found that 5 d.f. for cardiovascular-related outcomes and 4 d.f. for respiratory outcomes minimized qAIC.

Stratified analyses

We conducted stratified analysis by patient age group (0–17, 18–64, 65+) and race/ethnicity (non-Hispanic White vs. Hispanics) to investigate effect measure modifications. Black and Asian/Pacific Islander patients were not included because of small case numbers for these groups at the cluster-day level. Datasets stratified by the corresponding age or race/ethnicity categories were fitted to the two-stage model described above. Cardiovascular-related outcomes for patients aged 0–17 years were not examined due to insufficient data.

Furthermore, effect measure modification by cluster level poverty was investigated as a proxy for neighborhood SES. We used Jenks/natural breaks to classify the cluster-level “average percentages of reported household income of less than two times the federal poverty level” into three levels (low (<24%), moderate (24–<38%), and high ($\geq 38\%$), so that each level has approximately equal sized populations. Tests for interaction where we take the ratio of the difference in risk ratios to their standard errors were performed for all stratified analyses to quantify the degree of heterogeneity in estimates (Altman 2003).

Sensitivity analyses

Potential confounding by co-pollutants were examined. For each $PM_{2.5}$ metric examined in our primary model, we

included NO₂ and O₃, respectively, to examine the robustness of the main effects. Co-pollutants were matched to the PM_{2.5} exposures in metric, lag, and linearity; meaning daily maximum PM_{2.5} was fit with daily maximum NO₂ or O₃. PM_{2.5} DECH-9, -12, and -15 were fit with NO₂ DECH-5 and DECH-13 (ppb•hour) or O₃ DECH-30 (ppb•hour). Thresholds were selected based on the 2021 WHO Global

Air Quality Guideline for these pollutants (WHO 2021), which is 10 and 25 µg/m³ (5 and 13 ppb) for NO₂ and 60 (30 ppb) µg/m³ for O₃. Co-pollutants assumed the same lag period as PM_{2.5} and a linear relationship.

Results

Summary statistics

The final study population consisted of 35 regional clusters, with a 35,000 minimum population constraint for each cluster, constructed from the 455 ZCTAs in the South Coast AQMD domain (Supplemental Fig. 1). Each cluster covered 2 to 55 ZCTAs and has an average population of 477,655. Table 1 summarized the sociodemographic characteristics of the population in the study area at the cluster level. On average, 6.5% of residents reported non-Hispanic Black race, 46.8% reported Hispanic ethnicity, and 32.9% of households reported incomes of less than two times the federal poverty level. The population-weighted cluster median age was 36.8 years old.

Each cluster had 699 unique days of exposure and outcome data, from May of 2018 to March of 2020. The population weighted means (standard deviation) of DECH-9, DECH-12, DECH-15, daily maximum, and daily average PM_{2.5} level were 89.3 (106.7) µg/m³•hour, 56.7 (88.7) µg/m³•hour, 35.8 (72.2) µg/m³•hour, 19.3 (13.9) µg/m³, and 11.2 (5.6) µg/m³, respectively. Subdaily and daily metrics of O₃ and NO₂ levels were also calculated. Zero DECH values were observed on many days, specifically, 6.5%, 15.8%, and 26.5% of observations across all clusters for PM_{2.5} DECH-9, -12, and -15, 2.5% for O₃ DECH-30, and 1.4% and 17.3% for NO₂ DECH-5 and -13, respectively. Concentrations of PM_{2.5}, NO₂, O₃, and apparent temperature were weakly correlated (Spearman correlation coefficients < 0.4).

The average daily counts of cardiovascular- and respiratory-related EDVs were 4.9 (2.2) and 10.7 (6.4) per 100,000 population per cluster, respectively. When examining trends over time, cardiovascular-related EDVs showed little variation in distribution over the study period, whereas respiratory EDVs peaked during colder months.

Associations between daily EDVs and subdaily exposure metrics

The associations between daily EDVs and PM_{2.5} metrics were quantified as excess risk per interquartile range (IQR) increase in each exposure metric (Table 1). Figure 1 showed risk estimates of cardiovascular- and respiratory-related EDVs for each subdaily PM_{2.5} metric at different lags.

Table 1 Summary statistics of cluster-level demographics, outcomes, and exposures

Demographics	By cluster	
	Mean (SD) ²	Median (Min, Max)
# of ZCTAs (<i>N</i> =455)	13 (11.8)	10 (2, 55)
Population Count	477,655 (397,638)	383,934 (40,258, 1,531,522)
% Black	6.5% (8.1%)	4.9% (0.5%, 59.6%)
% Hispanic	46.8% (22.1%)	49.1% (15.6%, 92.3%)
% population with household income less than 2x FPL	32.9% (12.6%)	36.5% (17.0%, 58.6%)
Median age (years)	36.8 (4.2)	36.2 (29.7, 53.6)
Health Outcome(s) (per 100k population)	By Cluster-day	
	Mean (SD)	Median (Min, Max)
# of Cardiovascular-related EDVs	4.9 (2.2)	4.5 (0.4, 25.8)
# of Respiratory-related EDVs	10.7 (6.4)	9.0 (0.5, 54.1)
Exposure(s)	Mean (SD)	IQR
PM _{2.5}		
DECH-9 (µg/m ³ •hour)	89.3 (106.7)	110
DECH-12 (µg/m ³ •hour)	56.7 (88.7)	65.8
DECH-15 (µg/m ³ •hour)	35.8 (72.2)	32.7
Daily (24-hour) maximum (µg/m ³)	19.3 (13.9)	10.4
Daily (24-hour) average (µg/m ³)	11.2 (5.6)	6.8
Apparent Temperature (°F)	63.6 (10.7)	16.4
O ₃		
DECH-30 (ppb•hour)	171.8 (137.6)	180.1
Daytime 8-hour maximum ¹ (ppb)	46.0 (13.8)	16.1
Daily (24-hour) average (ppb)	29.9 (9.8)	13.7
NO ₂		
DECH-5 (ppb•hour)	179.8 (154.1)	212.7
DECH-13 (ppb•hour)	79.9 (104.1)	111.9
Daily (24-hour) maximum (ppb)	21.9 (10.1)	14.9
Daily (24-hour) average (ppb)	12.3 (6.9)	9.7

¹ Maximum of 8-hour average O₃.

² Population weighted mean and standard deviation were taken except # of ZCTAs, population count, and health outcomes, which is estimated by mean and standard deviation.

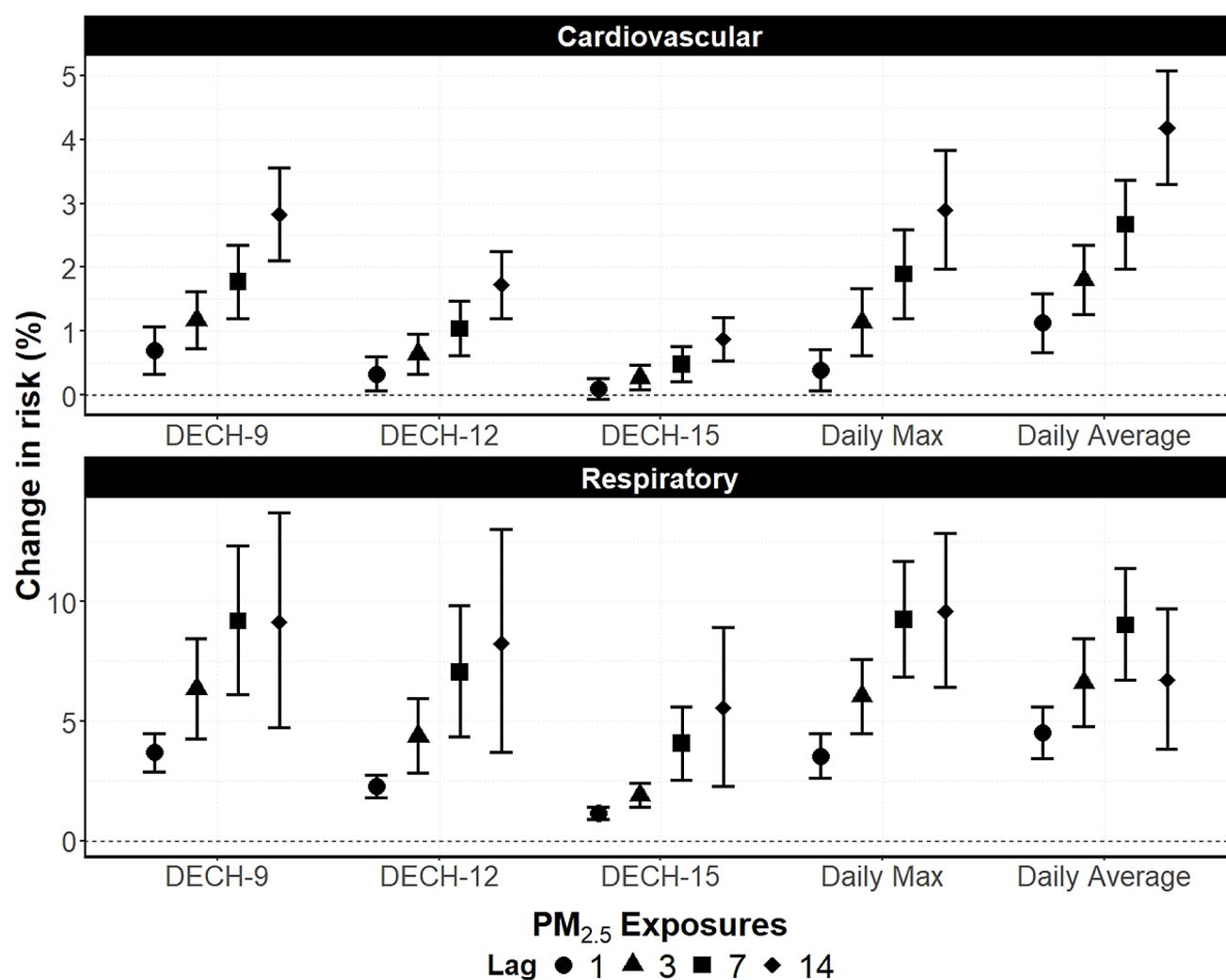


Fig. 1 Excess risk of cardiovascular- and respiratory-related EDVs and confidence intervals associated with an IQR increase in $PM_{2.5}$ exposure metrics at different distributed lags (exposure during the prior 1, 3, 7, 14 days to an EDV), Southern California (2018–2020)

The daily average concentration metric for $PM_{2.5}$ was also included for reference.

All subdaily $PM_{2.5}$ metrics were associated with elevated risk of cardiovascular-related EDVs, consistent with the daily $PM_{2.5}$ metric results (Fig. 1). For example, risk for cardiovascular-related EDVs increased by 1.77% (95% CI: 1.20, 2.34), 1.04% (0.61, 1.47), and 1.89% (1.21, 2.59) per IQR increase of DECH-9, DECH-12, and daily maximum $PM_{2.5}$ during 7-day lag period, respectively. Of the subdaily metrics, daily maximum $PM_{2.5}$ and DECH-9 showed greatest excess risks, most closely mirroring the associations for daily average $PM_{2.5}$. The associations for DECH-12 and DECH-15 were noticeably attenuated compared to DECH-9, possibly due to a large number of zero or near-zero values, as mentioned earlier.

Similarly, all subdaily $PM_{2.5}$ metrics were positively associated with respiratory EDVs across all lags. For example, risk of respiratory EDVs increased by 6.34%

(4.25, 8.48), 4.39% (2.85, 5.95), 1.91% (1.41, 2.41), 6.04% (4.51, 7.60), and 6.61% (4.78, 8.47) per each IQR increase of DECH-9, DECH-12, DECH-15, daily maximum $PM_{2.5}$ and daily average $PM_{2.5}$ during 3-day lag period. Furthermore, in contrast with the daily exposure metric, where excess respiratory EDV risk peaked with distributed 7-day lag but attenuated with 14-day lag, risk estimates of subdaily metrics remained or slightly increased with the longer lag than 7-day. This may suggest that the elevated risks from subdaily exposure on the respiratory system last longer than that of the daily metric.

Out of distributed lags tested, models with 7-day lags and 3-day lags had minimal qAIC for cardiovascular- and respiratory-related outcomes, respectively. Therefore, we present results of lag 7 for cardiovascular-related EDVs and lag 3 for respiratory EDVs for the following stratified and sensitivity analysis.

Stratified analyses

Stratification by age group

When stratifying by patient age, older adults (ages 65+ years) showed higher excess risks for cardiovascular-related outcomes across different $PM_{2.5}$ metrics compared to adults (ages 18–64 years), though differences were not statistically significant (Fig. 2).

For respiratory outcomes, children exhibited higher excess risks than both adult groups across all metrics. Differences were statistically significant for DECH metrics in tests for heterogeneity but not for the daily average $PM_{2.5}$ and daily maximum $PM_{2.5}$, though the effect sizes of DECH metrics were smaller than those of daily average

and maximum $PM_{2.5}$. Older adults also exhibited higher risks than adults; the differences were significant for $PM_{2.5}$ DECH-9, DECH-12, and daily average.

Again, for both types of outcomes, daily maximum and DECH-9 estimates resembled the association for daily average metric more closely. DECH-12 and DECH-15 showed smaller excess risk estimates, partially due to large percentage of zeros among DECH-12 and –15 values.

Stratification by race/ethnicity

We observed a slightly higher excess risk of cardiovascular-related EDVs among non-Hispanic White populations and a slightly higher excess risk of respiratory EDVs among Hispanic populations, although neither modification was

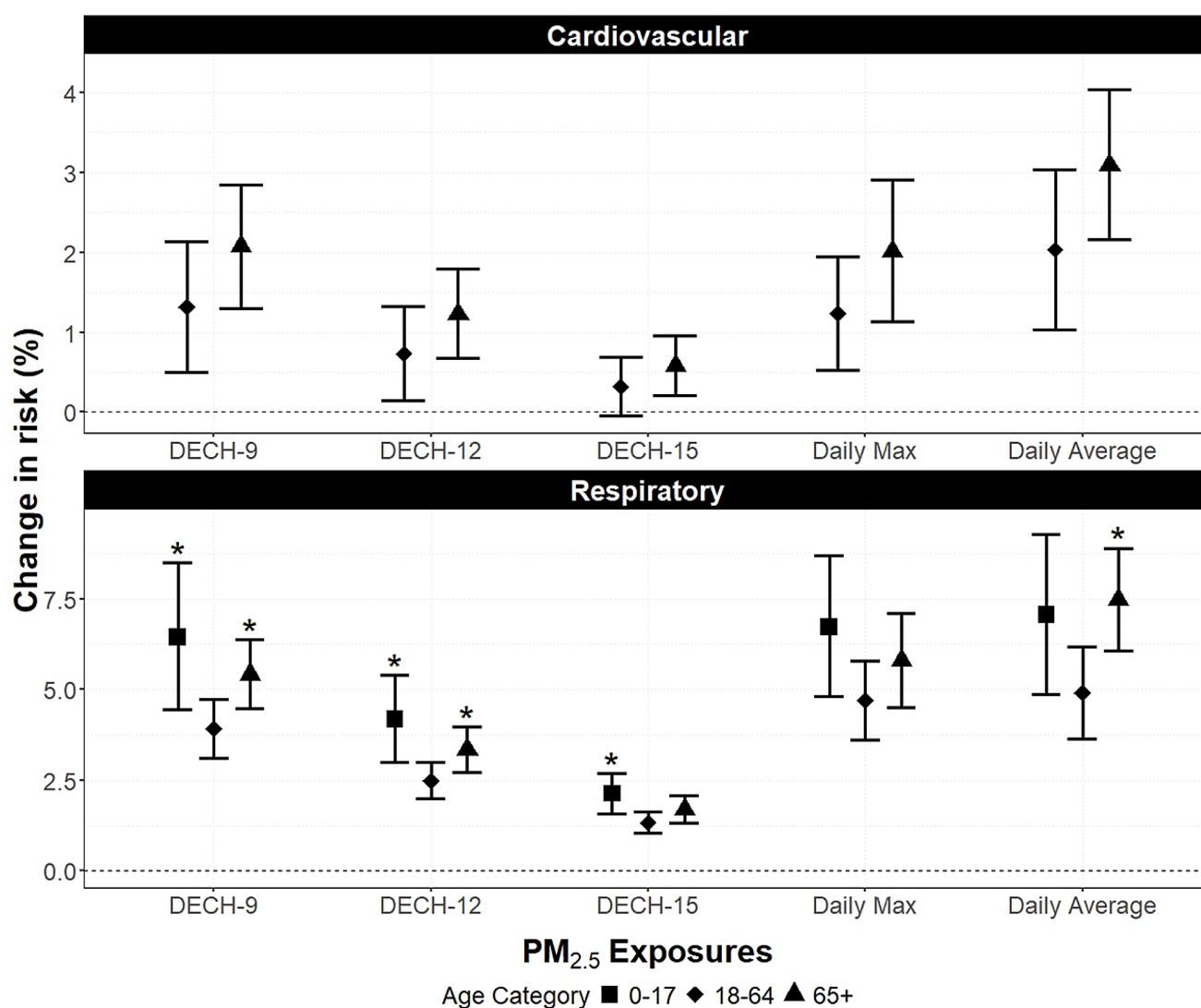


Fig. 2 Excess risk and confidence intervals of cardiovascular- and respiratory-related EDVs per each IQR increase of $PM_{2.5}$ exposure metrics, stratified by patient age group. Asterisk (*) represents statistically significant difference compared to the reference group, age

18–64. For cardiovascular-related outcomes, results of a 7-day lag were presented; the children category was not included given the low outcome counts. For respiratory outcomes, results of a 3-day lag were presented

statistically significant (Fig. 3). The variation in the risk estimates of respiratory EDVs between the non-Hispanic White and Hispanic group may be partly explained by the average ages of patients between strata (43.3 years for non-Hispanic White vs. 23.4 years for Hispanic). In contrast, the average ages of cardiovascular-related patients were similar between the two race/ethnicity groups. The direction and patterns of effect measure modification were consistent across subdaily metrics, as well as daily average $PM_{2.5}$.

Stratification by cluster poverty level

Given the strong modification effect by age group, the poverty stratification was conducted at the cluster level in addition to the age group stratification for each cluster (double

stratification). We divided clusters into three poverty-based strata: low (N of clusters=9, on average 19.3% of population with household income less than two times of federal poverty line); moderate ($N=12$, 35.1%); high ($N=14$, 49.5%) (Supplemental Table 1). Low poverty clusters had the lowest percentages of residents reporting Black race ($3.8 \pm 1.6\%$) and Hispanic ethnicity ($25.1 \pm 8.8\%$) and the highest median age (40.1 ± 1.6 years old) among the three strata. High poverty clusters had the highest percentage of Black residents ($9.1 \pm 8.2\%$), Hispanic residents ($68.5 \pm 14.9\%$), and youngest median age (32.9 ± 2.9 years old).

For cardiovascular-related EDVs, both adult groups living in low poverty clusters had higher excess risks than the moderate and high poverty groups across all $PM_{2.5}$ metrics (Fig. 4). For respiratory EDVs, older adults

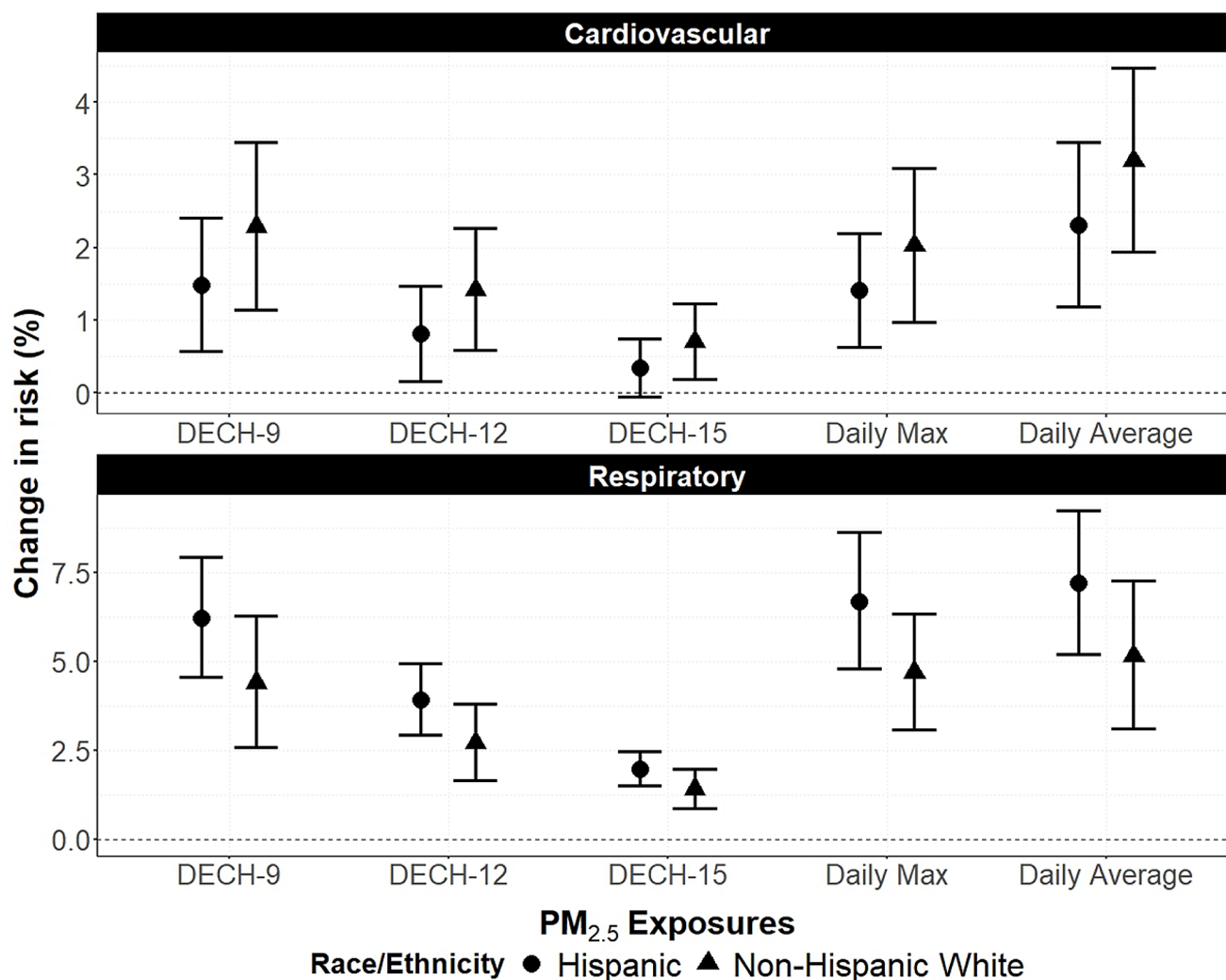


Fig. 3 Excess risk and confidence intervals for the association between $PM_{2.5}$ exposure metrics and EDVs per each IQR increase of $PM_{2.5}$ exposure metrics, stratified by patient race/ethnicity and outcome type. For cardiovascular-related outcomes, results of a 7-day lag were

presented; for respiratory outcomes, results of a 3-day lag were presented. None of the effect modification shown in this figure was statistically significant.

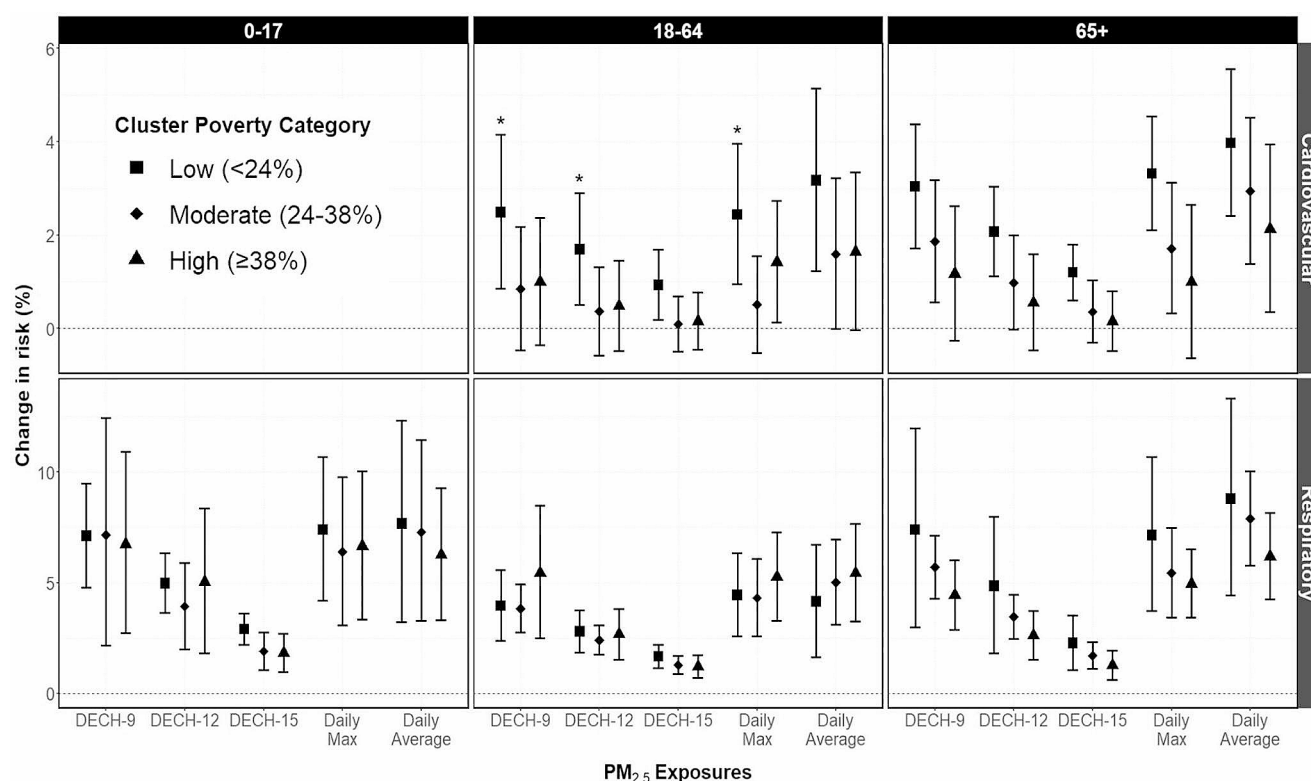


Fig. 4 Excess risk and confidence intervals for the association between $PM_{2.5}$ exposure metrics and EDVs per each IQR increase of $PM_{2.5}$ exposure metrics, stratified by patient age and cluster-level poverty. Asterisk (*) represents statistically significant difference compared to

the reference group, moderate poverty clusters. For cardiovascular-related outcomes, results of a 7-day lag were presented; for respiratory outcomes, results of a 3-day lag were presented

living in low poverty clusters showed higher excess risk estimates than those living in the moderate and high poverty strata but with wide CIs. As for children, excess risks based on DECH-15 suggested slightly higher risks of respiratory EDVs in the low poverty clusters, similar to the trend of daily average, but no clear variation was observed on daily max, DECH-9 or DECH-12 exposures. For adults (18–64 years), high poverty clusters showed slightly higher estimates of respiratory risks than the other two strata, as observed on DECH-9 and daily maximum, though these differences were not statistically significant.

These findings suggest the possible effect measure modification by poverty level on the relationship between $PM_{2.5}$ and cardiovascular-related outcomes for both adult groups and respiratory outcomes for children and older adults. Results based on subdaily metrics of $PM_{2.5}$ generally reflected the trend observed on daily average concentration, although the magnitude and direction of such modification differed by metric. Of the subdaily metrics, DECH-9 and daily maximum $PM_{2.5}$ produced estimates most similar to daily average $PM_{2.5}$, whereas DECH-12 and DECH-15 were more attenuated, mirroring the results of the primary model.

Sensitivity analyses

As a sensitivity test, co-pollutants, O_3 and NO_2 , were adjusted in the model (Fig. 5). $PM_{2.5}$ metrics were robust in the co-pollutant model with O_3 for both cardiovascular- and respiratory-related EDVs. When adjusting for NO_2 , the associations with cardiovascular-related EDVs remained stable. DECH-12 of $PM_{2.5}$ showed similar associations with cardiovascular-related outcomes when adjusted for DECH-5 and DECH-13 of NO_2 , compared to no adjustment. However, NO_2 appeared to attenuate the association between $PM_{2.5}$ and respiratory EDVs across all $PM_{2.5}$ metrics.

Discussion

We evaluated subdaily exposure to $PM_{2.5}$ and found that subdaily metrics of $PM_{2.5}$, daily maximum and DECH-9, generally showed similar elevated risks and patterns as daily average $PM_{2.5}$, the current $PM_{2.5}$ standard metric, on cardiorespiratory EDVs throughout our study period of 2018 to 2020 in Southern California. Subdaily metrics with higher thresholds (DECH-12 and –15) were comparatively attenuated. Patients in the older adult group consistently had

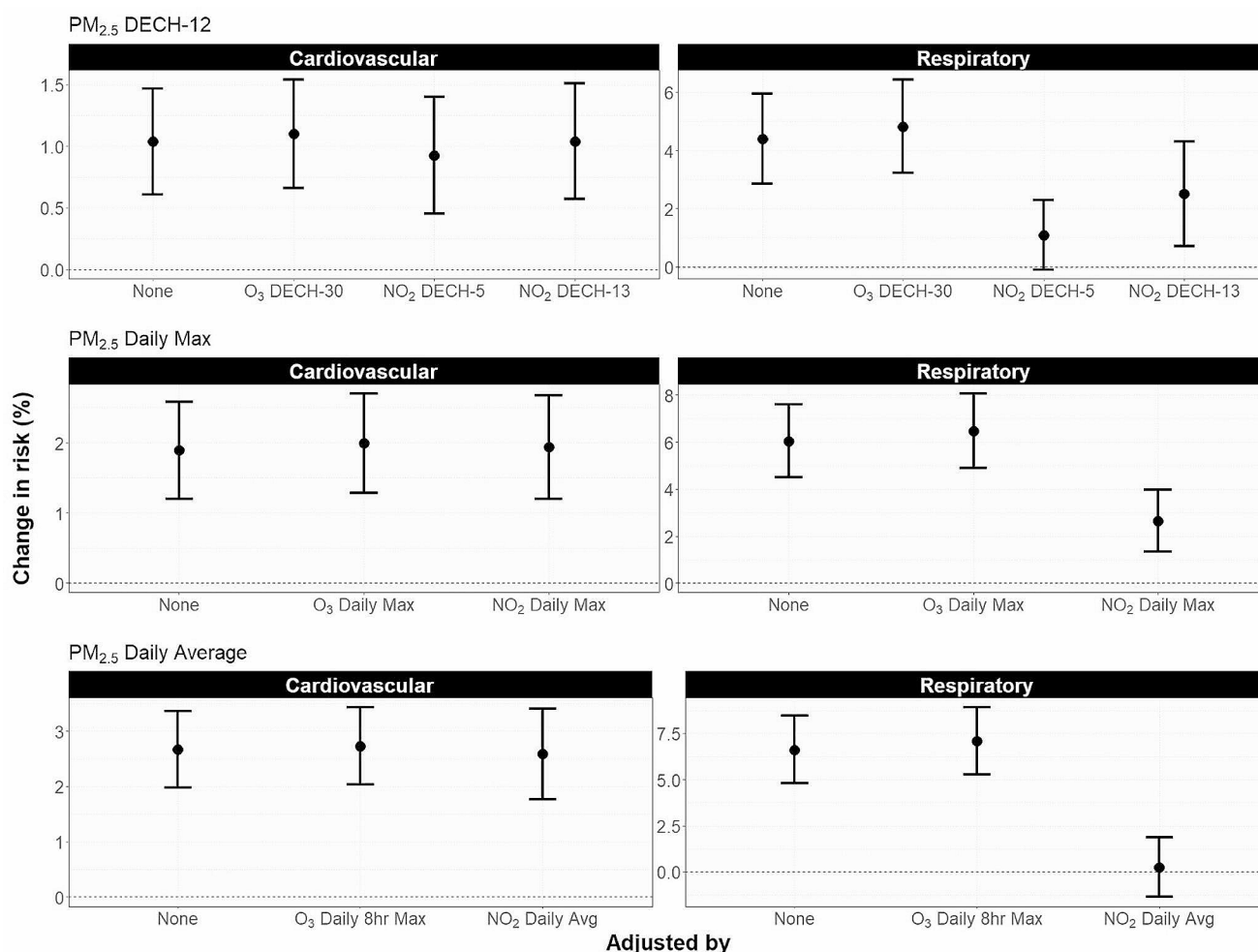


Fig. 5 Excess risks and confidence intervals for the association between $PM_{2.5}$ exposure metrics and EDVs per each IQR increase of exposure metrics, with and without co-pollutant adjustment. For cardiovascular-

related outcomes, results of a 7-day lag were presented; for respiratory outcomes, results of a 3-day lag were presented

a higher excess risk of cardiovascular-related outcomes, and the children and older adult groups showed higher excess risks of respiratory outcomes compared to adults. Race/ethnicity and neighborhood poverty level did not show consistent modification effect. Associations between $PM_{2.5}$ and cardiorespiratory outcomes were robust to the adjustment with O_3 and NO_2 , except that NO_2 attenuated the association of $PM_{2.5}$ and respiratory outcomes, indicating that traffic emissions may account for the majority of adverse respiratory outcomes.

Comparison to other studies

Our findings are in agreement with earlier research conducted outside of California, specifically, the associations of cardiorespiratory outcomes with subdaily $PM_{2.5}$ exposure were comparable or smaller than those with the daily average metric. Several studies in China have examined

the associations between DECH metrics and various cardiorespiratory outcomes, such as hospitalization of chronic obstructive pulmonary disease, outpatient visits of hypertension, and all-cause EDVs, reporting a similar reduction in effect when using DECH (Jiao et al. 2020; Lin et al. 2021; Zhang et al. 2021). However, another study looking into outpatient visits of acute lower respiratory infections among children reported stronger associations with DECH metrics than daily average $PM_{2.5}$ (Xiao et al. 2022). Such a difference could be due to the differences in study populations and medical systems, type and severity of health outcomes, and/or $PM_{2.5}$ sources and components. Therefore, results of studies in China may not be generalizable to California, which is one of the motivations of this study. One thing to note is that our data had many zero values for DECH-12 and -15, namely, the non-zero values of these DECH metrics only covered exposure of a portion of time period; whereas in most of the Chinese studies, DECH metrics had

non-zero values for almost the entire study periods, so the DECH results may be more close to the daily metric, such as DECH-9 in our study.

A number of studies using other subdaily metrics, such as daily maximum, hourly concentration, or exposure averaged across several hours, also obtained similar results. For example, the review by Buteau et al. (2019) found that the pooled odds ratios (ORs) of hospitalization for asthma and bronchiolitis in children < 5 years were 1.01 (95% CI: 0.98, 1.03) and 1.02 (0.93, 1.10) per 10 $\mu\text{g}/\text{m}^3$ increment in the hourly maximum concentration and daily mean of $\text{PM}_{2.5}$. A study in Israel using telemedicine data obtained similar ORs, 1.008 (95% CI: 0.998, 1.018) and 1.006 (0.995, 1.018), respectively, for any cardiac event associated with a 10 $\mu\text{g}/\text{m}^3$ increase in 6-h and 24-h average $\text{PM}_{2.5}$ (Nirel et al. 2018).

In this study, we examined the exposure up to a 14-day lag. Previous studies usually examined shorter lags and found that cardiovascular effects tended to be most influential on the same day or immediate lag periods (up to two days) (USEPA 2019); respiratory effects appeared more transient, with largest effect size observed within the hour of exposure and lagged up to 48 h. There was very limited evidence on associations for longer lags, i.e., 7-day or 14-day lags. Kim et al. (2012) observed strongest cardiovascular effects on the day 0 and another bump of risks around the lag 7th day, then the effect diminished thereafter. Along with this evidence, we observed cumulative exposures over longer lags were associated with greater risk increments in cardiovascular-related outcomes across all metrics. Respiratory risks associated with daily average $\text{PM}_{2.5}$ also accumulated along lags, peaking at lag 7 days. Such trends were also observed in earlier California studies (Bi et al. 2020; Delfino et al. 2014). We further found that, the respiratory risks associated with subdaily metrics remained after lag 7 days, compared to the decreasing trend of daily average $\text{PM}_{2.5}$, implying prolonged respiratory-related effects that were not reflected by daily average metric.

Health disparity is another focus of this study. Previous work in the U.S. have suggested cardiorespiratory health disparities with varying evidence that Black and Hispanic groups have higher exposure to air pollution and/or increased cardiorespiratory risks (Bell and Ebisu 2012; Boing et al. 2022; Liu et al. 2021), although some found higher $\text{PM}_{2.5}$ -associated respiratory hospitalization rate among White individuals compared to the overall population (Mehta et al. 2021). We observed higher excess risks of respiratory EDVs and lower excess risks of cardiovascular-related EDVs among Hispanic population associated with both subdaily and daily $\text{PM}_{2.5}$ metrics, compared to non-Hispanic White, likely driven by the age difference between the two groups. With a younger average age, the Hispanic

group contains a higher proportion of children vulnerable to respiratory outcomes. Furthermore, differing from previous studies, we observed more cardiovascular-related EDVs among people living in higher SES areas, for both adult groups. This might be attributed to multiple reasons: better health insurance coverage for people of higher SES; language and transportation barriers to access healthcare among lower SES groups; or lack of sick leave to seek medical care especially for the low-income workers (Peipins et al. 2012).

There are several established pathophysiological pathways that would explain the positive associations between $\text{PM}_{2.5}$ and the cardiorespiratory outcomes that we observed. Inhaled $\text{PM}_{2.5}$ can penetrate to the deep lung and enter the bloodstream, causing not only respiratory tract inflammation but also systematic inflammation responses and oxidative stress that may further trigger asthma and alter hemostasis which may also lead to MI, and heart failure, increase heart rate variability, coronary heart diseases, and increased blood pressure (USEPA 2019). Pulmonary inflammation and decreases in lung functions were largely observed as well, although these effects are considered transitory (USEPA 2019).

Implications of utilizing subdaily $\text{PM}_{2.5}$ metrics

The observed elevated risks for cardiorespiratory outcomes with exposure to $\text{PM}_{2.5}$ DECH-9 lend support for the new U.S. national ambient annual $\text{PM}_{2.5}$ standard values and adds to the body of evidence that indicates significant health effects below 12 $\mu\text{g}/\text{m}^3$. $\text{PM}_{2.5}$ exposure above 9 $\mu\text{g}/\text{m}^3$ shows similar risks of cardiovascular- and respiratory-related EDVs associated with the daily average metric.

In our findings, both daily maximum and DECH metrics readily follow the daily average metric, which aligned with USEPA's early determination. Daily average metric appeared to cover most of the cardiorespiratory-related effects suggested by the daily maximum and DECH metrics. Elevated subdaily transient exposures like fires and fireworks were not frequent enough to influence the overall associations in our study area, but the particular $\text{PM}_{2.5}$ components captured by DECH-12 and -15 may be more toxic than those in the regional background or more toxic above certain threshold, showing longer lagged respiratory effects. In this manner, subdaily measurements of $\text{PM}_{2.5}$ would be relevant in historically highly polluted areas and in the event of air pollution accidents. However, given the confounding effects by NO_2 , such prolonged effects need to be further confirmed.

Based on the existing results, the daily average metric was sufficient to characterize ambient $\text{PM}_{2.5}$ exposure and associated cardiorespiratory health outcomes, and health

guidance based on daily metric could provide sensible protection to the public in general. The hourly Air Quality Index (AQI) appears to be an easy and efficient way to communicate with the public, as hourly AQI is calculated by referencing hourly $PM_{2.5}$ and annual/daily National Ambient Air Quality Standards. Nevertheless, given the inconclusive finding, we recommend that both sensitive populations as well as healthcare providers to stay vigilant for elevated subdaily exposure and prepare for potential cardiorespiratory effects in their localities, particularly if they live in or provide care in polluted areas. Future studies in other densely populated areas, allowing for more socioeconomic stratifications and investigation of specific diseases, are needed to verify our findings, whereas conducting similar analysis in less populated areas would be challenging.

Strengths and limitations

Our study has many strengths. We utilized relatively complete, hourly gridded exposure data for a large portion of Southern California to create subdaily metrics. We were able to control for co-pollutants to parse out the individual effect of $PM_{2.5}$ on cardiorespiratory outcomes. Despite the challenge of low EDV counts at ZCTA level, we were able to leverage a spatial clustering procedure to create larger geographic units of observations based on sociodemographic likeness and exposure variability to increase outcome counts, especially in rural areas. This regionalization method may be a valuable tool for future analyses involving sparse counts and spatial autocorrelation of observations, making it possible to investigate potential effect measure modification by sociodemographic factors.

On the other hand, our analyses have several limitations. First, the EDV records came with date but no exact admission time stamp, which limited the scope of analysis to the daily level and the subdaily metrics we can use. Thus, we were not able to identify which particular hour(s) of exposure were most relevant. Second, the EDV counts at each geographic unit were small, even after aggregating to the cluster level, with many zeros in sparsely populated areas. These factors restricted our ability to look at specific health outcomes and stratification analysis by potential confounders. Several Chinese studies in recent years have reported specific types of diseases, such as ST-segment-elevation MI and acute lower respiratory infections, associated with transient exposure at hourly scale (Chen et al. 2022; Jiang et al. 2023; Lei et al. 2022). However, we have neither the health data specified to hourly resolution nor enough cases to look at specific types of diseases. Considering that the Los Angeles metropolitan area is one of most populated areas in U.S., future studies may limit their study populations to urban areas or target more populated areas, e.g., New York City, to

achieve higher case counts, but the demographics of these areas may not be representative of the U.S. general population. Third, there was likely exposure misclassification due to the regionalization process. While we attempted to maximize $PM_{2.5}$ similarity within-clusters, we lost geographic variation to some degree. This issue is common when aggregating information across geographic units of varying sizes and arbitrary administrative borders (Parenteau and Sawada 2011). However, the exposure misclassification was likely non-differential (random and not systematic) in our dataset, and therefore our estimates would be biased toward the null.

Conclusions

Our findings confirmed that subdaily exposures to elevated ambient air pollution were associated with increased adverse cardiorespiratory outcomes that were basically captured by the daily average metric. Peak subdaily exposure may have respiratory effects over an extended period, longer than the daily average metric implied, but the effect estimate was still within the quantification of the daily average metric. Further studies to consider prolonged risks and specific health outcomes are warranted, though having sufficient case counts for fine spatial resolution may be challenging.

With both subdaily and daily metrics, we observed elevated excess cardiovascular risks among older adults and elevated excess respiratory risks among both older adults and children compared to adults aged 18–64 years. We also observed excess risks of cardiorespiratory EDVs associated with ambient $PM_{2.5}$ exposure among low-poverty neighborhoods. More research is needed to investigate these associations and uncover the underlying reasons.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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Author contributions Xiangmei Wu and Keita Ebusu conceived of the presented idea and designed the study. Nico Schulte and Scott Epstein generated the hourly air data, and Keita Ebusu processed the health outcome data. Andrew Nguyen performed data analysis, particularly designed the regionalization process with the supervision of Keita Ebusu and Xiangmei Wu. The first draft of the manuscript was written by Andrew Nguyen, Xiangmei Wu, and Keita Ebusu. The draft was reviewed and edited by Keita Ebusu, Nico Schulte, Rupa Basu, Scott Epstein, and Xiangmei Wu. All authors commented on previous versions of the manuscript and approved the final manuscript.

Data availability The hourly PM_{2.5} and ozone concentration data used in this manuscript are available with a Public Records Request at. The emergency department visit data are available through the California Department of Health Care Access and Information with an approved application.

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